

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

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COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour
Total Marks: 30

Annexure A

Dry Run Evaluation

Code of Conduct:

I Preethi R (192511352) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R. Preethi

20528 + 19519 + 9
= 95
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Student 25: Preethi R | Register Number: 192511352

1. Question 23 - Agent Architecture

A self-driving car receives input from Camera, GPS, LiDAR and Speed Sensor and controls Steering, Brake and Accelerator.

Identify sensors, actuators, environment and percepts. Draw the basic agent structure and justify a suitable agent type.

2. Question 24 - CSP Exam Timetable

Schedule AI, ML, DBMS and OS into Monday 10 AM, Monday 2 PM and Tuesday 10 AM.

Constraints: AI and ML cannot share a slot; DBMS after AI; OS not Monday 2 PM; ML before OS.

Use backtracking and show every assignment.

Subject	Domain	Assigned Slot
AI	Mon10/Mon2/Tue10	
ML	Mon10/Mon2/Tue10	
DBMS	Mon10/Mon2/Tue10	
OS	Mon10/Mon2/Tue10	

3. Question 25 - Search Strategy Comparison

Consider a graph with $S \rightarrow A, B$; $A \rightarrow C, D$; $B \rightarrow E$; $C \rightarrow G$; $D \rightarrow G$; $E \rightarrow G$.

Perform BFS, DFS, UCS, Greedy Best-First Search and A* separately. Complete the comparison table.

Algorithm	Expansion Order	Path	Cost	Complete?	Optimal?
1					
2					
3					
4					
5					

23. Agent Architecture Dry Run

Given:

Sensors: Camera, GPS, LIDAR, Speed Sensor

Actuators: Steering, Brake, Accelerator

Environment: Road, vehicle, pedestrian, traffic signals, obstacles.

Steps	Sensor Input	Agent Decision	Actuator
1.	Camera detects traffic signal	Identify signal status	-
2.	LIDAR detects obstacle	Calculate obstacle distance	Brake
3.	GPS gives current location.	Determine route	Steering
4.	Speed Sensor gives current speed	Check whether speed is safe	Accelerator / Brake
5.	All Sensor data combined	Selects best action	Steering / Brake / Accelerator

Environment → Sensors → Agent → Actuators → Environment

Model-Based Reflex Agent because the car uses current sensor information along with an internal model / state of the environment to make decision.

24. CSP Exam Timetable.

Variables

- AI
- ML
- DBMS
- OS

Constraints

- 1) $AI \neq ML$
- 2) DBMS after AI
- 3) $OS \neq \text{Mon 2PM}$
- 4) ML before OS

Domains

- Mon 10AM
- Mon 2PM
- Tue 10AM

Backtracking:-

We assign variable one by one

Step: Assign AI

Try: AI Mon 10AM

No constraint violated

Step 2: Assign ML

Try: ML = Mon 10AM

But:

AI = Mon 10AM

ML = Mon 10AM

X Violates $AI \neq ML$

Back track.

Try:

ML = Mon 2PM

Valid.

AI = Mon 10AM

ML = Mon 2PM

Step 3: Assign DBMS

Constraint: DBMS after AI

Try: DBMS = Mon 10AM

X Not after A1

Try: DBMS = Mon 2PM

X Already occupied by ML

Try: DBMS = Tue 10AM

A1 = Mon 10AM

ML = Mon 2 PM

DBMS = Tue 10AM

Step 4: Assign OS

Constraint:

→ OS ≠ Mon 2PM

→ ML before OS

Since ML = Mon 2PM, OS must be after Mon 2PM.

Try:

OS = Tue 10AM

But DBMS is already there.

Therefore this branch fails

Backtrack.

We need to change an earlier assignment.

Change ML:

ML = Tue 10AM

But DBMS must be after A1, and available slot after A1 is Mon 2PM / Tue 10AM.

Try:

DBMS = Mon 2PM

But DBMS must be after

Then:

AI = Mon 10 AM

DBMS = Mon 2 PM

ML = Tue 10 AM

25. Search Strategy Comparison

Given Graph:

$S \rightarrow A, B$

$A \rightarrow C, D$

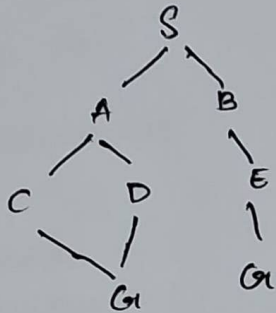
$B \rightarrow E$

$C \rightarrow G$

$D \rightarrow G$

$E \rightarrow G$

We assume the child order is left to right



BFS - Breadth first Search.

BFS uses a Queue.

Dry run Start:

Queue = [S]

Expand S:

Queue = [A, B]

Expand A:

Queue = [B, C, D]

Expand B:

Queue = [C, D, E]

Expand C:

Queue = [D, E, G]

Expand D:

Queue = [E, G]

Expand E:

Queue = [G]

G is reached.

Expansion order: $S \rightarrow A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow G$

path: $S \rightarrow A \rightarrow C \rightarrow G$

Cost: All edge assumed unit cost: 3

DFS - Depth first Search

DFS uses a stack

Start at S and go as deep.

$S \rightarrow A \rightarrow C \rightarrow G$

Goal reached

Expansion order: $S \rightarrow A \rightarrow C \rightarrow G$

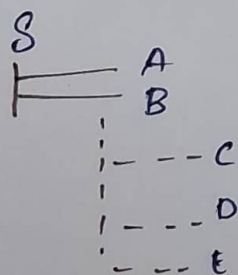
path: $S \rightarrow A \rightarrow C \rightarrow G$

Cost: 3

UCS - Uniform Cost Search:

Assume each edge cost = 1

UCS always chooses the node having the lowest path cost $g(n)$.



$g = 0$

$g = 1$

$g = 1$

$g = 2$

$g = 2$

$g = 2$

Dry run: $S \rightarrow A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow G$

Goal G & reached with cost:

$$g(G) = 3$$

path: $S \rightarrow A \rightarrow C \rightarrow G$

Greedy Best-first Search

Greedy uses only: $f(n) = h(n)$

Since the question does not provide heuristic values, assume

Node	$h(n)$
S	3
A	2
B	2
C	1
D	1
E	1
G	0

Start: S from S

A ($h=2$), B ($h=2$)

Choose A.

from A:

C ($h=1$), D ($h=1$)

Choose C

from C:

G ($h=0$)

Choose G

Expansion order: $S \rightarrow A \rightarrow C \rightarrow G$

path: $S \rightarrow A \rightarrow C \rightarrow G$

Cost: 3

A* Search

A* Uses:

$$f(n) = g(n) + h(n)$$

using the same heuristic:

Node	$g(n)$	$h(n)$	$f(n)$
S	0	3	3
A	1	2	3
B	1	2	3
C	2	1	3
D	2	1	3
E	2	1	3
G	3	0	3

Dry run: $S \rightarrow A \rightarrow C \rightarrow G$

Expansion order: $S \rightarrow A \rightarrow C \rightarrow G$

Cost: 3



RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	20	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	30	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	20	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	20	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	10	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

$$20 + 28 + 19 + 19 + 9$$

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